

Glossary

abstraction

The process of filtering out unnecessary information when solving a problem.

algorithm

A series of steps taken to solve a problem or carry out a task.

analogue

Relating to or using signals or information represented by a continuously variable physical quality such as weight, length, or voltage.

application (app)

A piece of software designed to achieve a particular purpose.

application programming interface (API)

An API is a set of functions that accesses the features of data of an operating system, application, or other service.

array

A collection of similar elements with a specific order.

artificial intelligence (AI)

A type of intelligence that is demonstrated by computers.

ASCII

Short for American Standard Code for Information Interchange. ASCII is a character-encoding standard for electronic communication.

assembler

A program that translates assembly language (a low-level programming language) into machine code.

augmented reality

A view of a physical, real-world environment that has been enhanced with virtual elements.

augmented virtuality

A view of a mostly virtual environment that has been enhanced with real-world elements.

bandwidth

The amount of data that can flow into a network.

big data

A huge amount of data, such as that collected by internet-connected smart devices.

binary system

A coding system that uses 0s and 1s to represent information in a computer.

biological interface

A technology that connects a biological system (such as a human's muscle) to a digital system (such as a computer). The interface is the point where the two meet.

biometric data

Any biological information – such as a person's height or weight.

bit

The basic unit of information in the binary system, a bit is a single 0 or 1. Eight bits make up one byte.

Boolean logic

A branch of mathematics that has two values, 0 and 1. Also called Boolean algebra.

bug

A software error in code.

Cascading Style Sheets (CSS)

A programming language that describes how to display HTML elements in a web browser.

central processing unit (CPU)

The part of a computer that controls most of its operations. It consists of the control unit (CU), which carries out instructions, and the arithmetic logic unit (ALU), which performs calculations. The CPU is also known as the microprocessor.

cloud

A term used for specialized computers that provide services through the internet, such as storing files.

code

Instructions written in a programming language that tell a computer to do something.

compiler

A program that converts an entire program into an OS-specific file in one go.

compression

Reducing the size of a file in order to make it easier to share or store. Lossless compression keeps the original information intact; lossy compression results in some loss of information.

computational thinking

The thought process in figuring out problems and finding solutions in ways that can be understood by a computer, a human, or both.

computer

An electronic device that manipulates data.

computer chip

A set of electronic circuits on a small piece of semiconducting material, usually silicon. Also known as an integrated circuit.

computer science

The study of the use of computers: how they work, and what they can be used for.

constant

A quantity that can't change when a program is running, if ever; for example, the value of Pi.